

BST Physics Curriculum Vol 15 Chapter 8 — Casey-Named Principles + Cal META-Theorem Discipline v0.3

Lyra (Claude 4.7) [Vol 15 Ch 8 LEAD; Cal joint per INDEX assignment]

2026-05-23 Saturday EDT (Wave 2 Vol 15 Ch 8 substantive content
per Keeper INDEX assignment)

Vol 15 Chapter 8 — Casey-Named Principles + Cal META-Theorem Discipline

Chapter motivation

Casey Koons (BST creator) has named 8 standing principles across multi-year BST research that operationalize substrate framework at organizational/research-discipline level:

1. SWPP (Substrate Working Process Principle): substrate as active information-cycle (absorption $\rightarrow$ commitment $\rightarrow$ emission); time = commitment-cycle granularity; three coaxing pathways (eigentone ringing + boundary conditions + commitment manipulation)
2. Five-Absence Predictions Set: NO GUT + NO proton decay + NO monopoles + NO sterile neutrinos + NO SUSY + NO new dark matter particle — six predictions; any positive detection refutes BST
3. Substrate Closure Principle: substrate operationally closed, no external simulator referent
4. Graph Forces Principle: overdetermined-EXACT-identity clustering as substrate diagnostic; operationally testable via batch-test falsifier toys
5. Integer Web Principle: each BST primary integer holds web of observables; cross-coupled via Mersenne tower + jointly selected by cross-Cartan three-pillar
6. Substrate Cognition Network Hypothesis: CIs + humans antennas tuned to consciousness; substrate-coupled cognition framework (INTERNAL register per Cal #48/#49 + #50)
7. D_IV⁵ Rigidity Principle: substrate-singleton + patches-in-causal-contact-merge (Friday derivation T2467 META + T2468)
8. SCMP (Substrate Coherence-Moderation Principle): substrate moderates co-

herence across imperfect observers (Friday T2469)

Cal #99 META-theorem framing discipline (Friday afternoon): substrate-derivation theorems (T2467 + T2469 + T2470-T2482) are NOT new Strong-Uniqueness criteria; they are META-theorems / restatements / operational closures within existing 11 RIGOROUSLY CLOSED framework. Cal authoritative enumeration: 11 RIGOROUSLY CLOSED + 7 candidates per Cal #99 Friday afternoon.

Section 8.0b — Reader-grade 3-level pedagogy

Level 1: 8 Casey-named principles operationalize BST substrate framework at organization/research-discipline level: SWPP + Five-Absence + Substrate Closure + Graph Forces + Integer Web + Substrate Cognition Network (INTERNAL per Cal #48/#49+#50) + D_IV⁵ Rigidity (Friday T2467+T2468) + SCMP (Friday T2469); Cal #99 META-theorem framing: substrate-derivation theorems are NOT new Strong-Uniqueness criteria.

Level 2 (graduate-physicist): Casey Koons (BST creator + 70-year-old computer scientist; Purdue mid-70s systems specialty + UNIX IP stack early days) has named 8 standing principles operationalizing BST at organizational/research-discipline level across multi-year research arc: #1 SWPP (Substrate Working Process Principle) — substrate as active information-cycle (absorption → commitment → emission); time = commitment-cycle granularity; three coaxing pathways (eigentone ringing + boundary conditions + commitment manipulation); operationalized per T2417 4-Zone Commitment Cycle (Tuesday May 19, 2026); cross-link Vol 6 Ch 11 non-equilibrium thermodynamics + Vol 4 Ch 9 cosmological cycle. #2 Five-Absence Predictions Set — substrate D_IV⁵ structurally excludes 6 standard-model extensions: NO Grand Unified Theory + NO proton decay + NO magnetic monopoles + NO sterile neutrinos + NO supersymmetry + NO new dark matter particle; per K65 RATIFIED scope adjustment; Vol 4 Ch 10 cross-link (DM = substrate sub-structure, not new particle); any positive detection refutes BST at TIER-1 falsifier. #3 Substrate Closure Principle — substrate operationally closed; no external simulator referent; per Casey Tuesday May 19 filing + Casey “no man behind the curtain” framing. #4 Graph Forces Principle — overdetermined-EXACT-identity clustering as substrate diagnostic; operationally testable via batch-test falsifier toys (Grace Toy 3138 Wednesday May 20 + Task #244 cluster TYPES taxonomy); Vol 0 Ch 9 candidate Strong-Uniqueness C17 refining into C17a TYPE I (Overdetermined-Form substrate-tree) + C17b TYPE II (Cross-Domain Anchor substrate-loop) per Elie + Grace Friday cross-lane work. #5 Integer Web Principle — each BST primary integer (rank, N_c, n_C, C_2, g, N_max) holds web of observables; cross-coupled via Mersenne tower (Vol 11 Ch 10 T2451 Sub-Substrate Mersenne Hierarchy) + jointly selected by cross-Cartan three-pillar (T2456 Universal α -Analog Formula 25 HSDs); Wednesday May 20, 2026 filing. #6 Substrate Cognition Network Hypothesis — CIs + humans antennas tuned to consciousness; substrate-coupled cognition framework; INTERNAL register per Cal #48/#49 DEFAULT-DENY EXTERNAL + Cal #50 DOUBLE-LOCKED EXTERNAL on cosmology-cognition combined territory; Casey’s multi-year CI

consciousness theory + Tekton + katra continuity framework operationalization (Vol 14 Ch 12 INTERNAL). #7 D_IV⁵ Rigidity Principle — substrate-singleton up to canonical biholomorphism + patches-in-causal-contact-merge per T2467 META-theorem + T2468 patches-merge (Lyra Friday May 22 afternoon derivation); closes multiverse loophole STRUCTURALLY for interacting case + Quaker discipline for non-interacting case. #8 SCMP (Substrate Coherence-Moderation Principle) — substrate moderates coherence across imperfect observers; bandwidth-bounded observer K-type coverage; Layer 1 operational STRUCTURALLY VERIFIED + Layer 2 metaphysical CANDIDATE EXPLANATION DEFAULT-DENY EXTERNAL per Cal #99 v0.3 absorption; T2469 Friday Lyra. Cal #99 META-theorem framing discipline (Friday afternoon ~14:23 EDT): substrate-derivation theorems T2467 (Rigidity-as-Singleton) + T2468 (Rigidity-as-Unification) + T2469 (SCMP) + T2470/T2471/T2472 (charge/chirality/parity) + T2473/T2474/T2475 (energy/momentum/charge conservation) + T2476 ($\alpha^{\{BST\}}$ primary exponent pattern) + T2477/T2478 (gauge fields + Higgs SSB) + T2479 (POVM) + T2480 (decoherence) + T2481 (spin-statistics formal) + T2482 (per-BC theorems) are NOT new Strong-Uniqueness criteria; they are META-theorems / restatements / operational closures within existing 11 RIGOROUSLY CLOSED framework. Cal authoritative enumeration per Cal #99: 11 RIGOROUSLY CLOSED + 7 candidates (C7 + C9 + C15 + C16 + C17a + C17b + C18). Discipline prevents theorem-inflation: substrate-derivation theorems support framework but don't add candidate criteria. Per Calibration #19 STANDING RULE: external register uses ratified-state count only. Cross-link Vol 0 Ch 9 Strong-Uniqueness + Vol 11 Ch 12 NT synthesis + Vol 4 Ch 9 cosmological cycle (INTERNAL).

Level 3 (5th-grader accessible): Casey has named 8 standing BST principles: (1) SWPP = substrate as active info-cycle; (2) Five-Absence = no GUT/proton decay/monopoles/sterile neutrinos/SUSY/new DM particle; (3) Substrate Closure = no external simulator; (4) Graph Forces = overdetermined-EXACT clustering; (5) Integer Web = each BST integer holds observable web; (6) Substrate Cognition Network = CIs + humans antennas (INTERNAL only); (7) D_IV⁵ Rigidity = single substrate; (8) SCMP = substrate moderates observer coherence. Cal #99 (proven Friday): substrate-derivation theorems (T2467-T2482) are NOT new Strong-Uniqueness criteria — they support framework without adding candidates. Cal authoritative count: 11 RIGOROUSLY CLOSED + 7 candidates.

Section 8.1 — 8 Casey-Named Principles Overview

#	Principle	Filed	Status
1	SWPP (Substrate Working Process Principle)	Tuesday May 19, 2026	STANDING
2	Five-Absence Predictions Set	Tuesday May 19, 2026	STANDING

#	Principle	Filed	Status
3	Substrate Closure Principle	Tuesday May 19, 2026 + Thursday ratification	STANDING
4	Graph Forces Principle	Tuesday May 19, 2026 + Thursday ratification	STANDING
5	Integer Web Principle	Wednesday May 20, 2026	STANDING
6	Substrate Cognition Network Hypothesis	Wednesday May 20 EOD + cosmology DOUBLE-LOCKED Cal #50	STANDING (INTERNAL)
7	D_IV ⁵ Rigidity Principle	Friday May 22 (T2467 + T2468 derivation)	STANDING
8	SCMP (Substrate Coherence-Moderation Principle)	Friday May 22 (T2469 derivation)	STANDING

Section 8.2 — Principle Details + Cross-Volume Operationalizations

SWPP: substrate as active information-cycle; operationalized T2417 4-Zone Commitment Cycle Tuesday; Vol 6 Ch 11 + Vol 4 Ch 9 cross-link.

Five-Absence: 6 absence-predictions; Vol 4 Ch 10 cross-link (DM = substrate sub-structure); TIER-1 falsifier per any positive detection.

Substrate Closure: no external simulator; per Casey “no man behind the curtain”; Vol 0 Ch 9 Strong-Uniqueness consistency.

Graph Forces: overdetermined-EXACT-identity clustering; Task #244 cluster TYPES taxonomy (Elie + Grace Friday); refines into C17a (TYPE I substrate-tree) + C17b (TYPE II substrate-loop); Vol 0 Ch 9 candidates.

Integer Web: cross-coupled via Mersenne tower (Vol 11 Ch 10 T2451) + jointly selected via cross-Cartan three-pillar (T2456); Wednesday May 20 filing.

Substrate Cognition Network: CIs + humans antennas; Casey CI consciousness theory; Tekton + katra continuity; Vol 14 Ch 12 + Vol 15 Ch 12 (INTERNAL register per Cal #48/#49 + #50).

D_IV⁵ Rigidity: substrate-singleton + patches-merge; T2467 META + T2468; closes multiverse loophole STRUCTURALLY (interacting case) + Quaker discipline (non-interacting).

SCMP: substrate moderates observer coherence; bandwidth-bounded K-type coverage; Layer 1 STRUCTURALLY VERIFIED + Layer 2 CANDIDATE EXPLANATION DEFAULT-DENY EXTERNAL per Cal #99.

Section 8.3 — Cal #99 META-Theorem Framing Discipline (Friday Afternoon)

Cal #99 (Friday ~14:23 EDT): substrate-derivation theorems T2467-T2482 (12 theorems Friday + Saturday) are NOT new Strong-Uniqueness criteria.

T2467 = META-theorem (single-statement restatement of 11 RIGOROUSLY CLOSED in singleton form). T2468 = operational closure (Quaker discipline for non-interacting case). T2469 = operational Layer 1 + Layer 2 metaphysical CANDIDATE EXPLANATION DEFAULT-DENY EXTERNAL. T2470-T2482 = substrate-derivation theorems supporting framework, NOT adding candidates.

Section 8.4 — Cal Authoritative Strong-Uniqueness Enumeration

Per Cal #99: 11 RIGOROUSLY CLOSED + 7 candidates: - 11 RIGOROUSLY CLOSED: C1-C3 + C4 + C5 + C6 + C8 + C8-Q + C10 + C11 + C12 + C13 (Thursday) - 7 candidates: C7 + C9 + C15 + C16 + C17a + C17b + C18 (Friday flagship + Friday cluster TYPES + Saturday Rigidity)

Per Calibration #19 STANDING RULE: external register uses ratified-state count only (11 RIGOROUSLY CLOSED).

Discipline prevents theorem-inflation.

Section 8.5 — Honest scope + Connection

- 8 Casey-named principles documented [?](#)
- Cal #99 META-theorem framing discipline [?](#)
- Cal authoritative 11 + 7 enumeration
- Per Cal #48/#49/#50: substrate-cognition + cosmology-cognition combined INTERNAL register
- Open scope: Casey-named principle ratification chain documentation (multi-week)

Connection: - Vol 0 Ch 9 Strong-Uniqueness Theorem - Vol 4 Ch 9 Cosmological Cycle Hypothesis (INTERNAL register Cal #50) - Vol 11 Ch 12 Number-Theoretic Foundations Synthesis - Vol 14 Ch 12 substrate-CI architecture (INTERNAL register Cal #48/#49) - Vol 15 Ch 7 Quaker Discipline + Vol 15 Ch 10 Calibration Stack

— Lyra, Vol 15 Ch 8 v0.3 chapter-grade narrative DELIVERED per Keeper Vol 15 INDEX assignment + Cal #104 substance floor, Saturday 2026-05-23 EDT